Project: Worked examples
Title: Load combinations (EN 1990)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

## 1. Scope and method

Design values of a single action effect (a force, a moment or a load intensity) under every EN 1990 Annex A1 combination for buildings: ULS persistent/transient, SLS characteristic, frequent and quasi-permanent, and accidental. Every variable action is tried as the leading one; the maximum and minimum envelopes and their governing combinations are reported.
ULS 6.10: Σ γ_G·G + γ_Q·K_FI·Q_1 + Σ γ_Q·K_FI·ψ_0·Q_i ULS 6.10a: Σ γ_G·G + Σ γ_Q·K_FI·ψ_0·Q_i; 6.10b: Σ ξ·γ_G·G + γ_Q·K_FI·Q_1 + Σ γ_Q·K_FI·ψ_0·Q_i Permanent factor: γ_G,sup when the action adds to the effect sought, γ_G,inf when it opposes it Variable actions: included only when they add to the effect sought (otherwise zero) SLS characteristic 6.14b: Σ G + Q_1 + Σ ψ_0·Q_i; frequent 6.15b: Σ G + ψ_1·Q_1 + Σ ψ_2·Q_i; quasi-permanent 6.16b: Σ G + Σ ψ_2·Q_i Accidental 6.11b: Σ G + A_d + ψ_1·Q_1 + Σ ψ_2·Q_i
Ref: EN 1990:2002+A1:2005 cl. 6.4.3.2 — Combinations for persistent or transient design situations (6.10, 6.10a, 6.10b)
Ref: EN 1990:2002+A1:2005 cl. 6.4.3.3 — Combinations for accidental design situations (6.11b)
Ref: EN 1990:2002+A1:2005 cl. 6.5.3 — Serviceability combinations (6.14b, 6.15b, 6.16b)
Ref: EN 1990:2002+A1:2005 Annex A1, Tables A1.1, A1.2(B) — Application for buildings: ψ and γ factors
Ref: UK NA to BS EN 1990:2002+A1:2005 NA.A1.2(A)–(C) — UK values of partial factors

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| actions | Actions (one effect, signed) | 2 row(s) | | drawing: GA-L2-001 rev C | any |
| actions[1] | | name = Slab, finishes and services; category = G permanent; F_k = 18 | | | |
| actions[2] | | name = Office imposed; category = B office; F_k = 10 | | | |
| unit | Unit of the effect | kN/m | | assumption: Line load on the beam. | any |
| method | ULS expression | 6.10a and 6.10b | | spec: UK NA Table NA.A1.2(B): the less favourable of 6.10a and 6.10b | any |
| gamma_G_sup | γ_G,sup (unfavourable permanent) | 1.35 | | spec: UK NA Table NA.A1.2(B) | 1 … 2 |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| gamma_G_inf | γ_G,inf (favourable permanent) | 1 | | spec: UK NA Table NA.A1.2(B) | 0.8 … 1.2 |
| gamma_Q | γ_Q (variable) | 1.5 | | spec: UK NA Table NA.A1.2(B) | 1 … 2 |
| xi | ξ reduction for 6.10b | 0.925 | | spec: UK NA Table NA.A1.2(B) | 0.8 … 1 |
| K_FI | K_FI consequence-class factor | 1 | | assumption: Consequence class CC2. | 0.9 … 1.1 |

## 3. Assumptions

A1. One scalar action effect; every action's characteristic value is its contribution to that effect, with sign. (Linear superposition (EN 1990 cl. 6.3.1 representative values).)
A2. Buildings only (Annex A1); no bridges, fatigue or seismic combinations. (EN 1990 Annex A1 scope.)
A3. Without a ψ record the EN 1990 Annex A1 recommended values are used, labelled on the sheet; the UK National Annex (NA.A1.2(A)–(C)) values come from a user record. (§A5: no NA table ships.)
A4. Each permanent action takes γ_G,sup or γ_G,inf by whether it adds to or opposes the effect sought; for EQU single-source checks, enter a source's permanent actions as one row. (EN 1990 Table A1.2(B) note on favourable and unfavourable permanent actions.)
A5. unit (Unit of the effect) is taken as kN/m. (Line load on the beam.)
A6. K_FI (K_FI consequence-class factor) is taken as 1. (Consequence class CC2.)
A7. ψ factors: EN 1990 Annex A1 recommended values (Table A1.1). (No ψ record entered.)

## 4. Working

[8] C1 = □_G,sup·G1 + □_Q·K_FI·□₀·Q2 — EN 1990 cl. 6.4.3.2
= 1.35×18 + 1.05×10
= 34.8 kN/m
ULS 6.10a (max)

[9] C2 = □·□_G,sup·G1 + □_Q·K_FI·Q2 — EN 1990 cl. 6.4.3.2
= 1.2488×18 + 1.5×10
= 37.478 kN/m
ULS 6.10b, Office imposed leading (max)

[10] C3 = 1·G1 + 1·Q2 — EN 1990 cl. 6.5.3
= 1×18 + 1×10
= 28 kN/m
SLS characteristic 6.14b, Office imposed leading (max)

[11] C4 = 1·G1 + □₁·Q2 — EN 1990 cl. 6.5.3
= 1×18 + 0.5×10
= 23 kN/m
SLS frequent 6.15b, Office imposed leading (max)

[12] C5 = 1·G1 + □₂·Q2 — EN 1990 cl. 6.5.3
= 1×18 + 0.3×10
= 21 kN/m
SLS quasi-permanent 6.16b (max)

[13] D1 = □_G,inf·G1 — EN 1990 cl. 6.4.3.2
= 1×18
= 18 kN/m
ULS 6.10a (min)

[14] D2 = ⊠_G,inf·G1 EN 1990 cl. 6.4.3.2
= 1×18
= 18 kN/m
ULS 6.10b, no variable action (min)
[15] D3 = 1·G1 EN 1990 cl. 6.5.3
= 1×18
= 18 kN/m
SLS characteristic 6.14b, no variable action (min)
[16] D4 = 1·G1 EN 1990 cl. 6.5.3
= 1×18
= 18 kN/m
SLS frequent 6.15b, no variable action (min)
[17] D5 = 1·G1 EN 1990 cl. 6.5.3
= 1×18
= 18 kN/m
SLS quasi-permanent 6.16b (min)
[18] E_d_max = maximum over the combinations
= ULS 6.10b, Office imposed leading
= 37.478 kN/m
ULS design effect, maximum
[19] E_d_min = minimum over the combinations
= ULS 6.10a
= 18 kN/m
ULS design effect, minimum
[20] E_k_max = maximum over the combinations
= SLS characteristic 6.14b, Office imposed leading
= 28 kN/m
SLS characteristic effect, maximum
[21] E_k_min = minimum over the combinations
= SLS characteristic 6.14b, no variable action
= 18 kN/m
SLS characteristic effect, minimum
[22] E_fr_max = maximum over the combinations
= SLS frequent 6.15b, Office imposed leading
= 23 kN/m
SLS frequent effect, maximum
[23] E_fr_min = minimum over the combinations
= SLS frequent 6.15b, no variable action
= 18 kN/m
SLS frequent effect, minimum
[24] E_qp_max = maximum over the combinations
= SLS quasi-permanent 6.16b
= 21 kN/m
SLS quasi-permanent effect, maximum
[25] E_qp_min = minimum over the combinations
= SLS quasi-permanent 6.16b
= 18 kN/m
SLS quasi-permanent effect, minimum

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| E_d_max | 37.478 kN/m | ULS design effect, maximum |
| E_d_min | 18 kN/m | ULS design effect, minimum |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| E_d_gov | "ULS 6.10b, Office imposed leading" | ULS design effect, governing (max) |
| E_k_max | 28 kN/m | SLS characteristic effect, maximum |
| E_k_min | 18 kN/m | SLS characteristic effect, minimum |
| E_k_gov | "SLS characteristic 6.14b, Office imposed leading" | SLS characteristic effect, governing (max) |
| E_fr_max | 23 kN/m | SLS frequent effect, maximum |
| E_fr_min | 18 kN/m | SLS frequent effect, minimum |
| E_fr_gov | "SLS frequent 6.15b, Office imposed leading" | SLS frequent effect, governing (max) |
| E_qp_max | 21 kN/m | SLS quasi-permanent effect, maximum |
| E_qp_min | 18 kN/m | SLS quasi-permanent effect, minimum |
| E_qp_gov | "SLS quasi-permanent 6.16b" | SLS quasi-permanent effect, governing (max) |

## 6. Checks (governing first)

No pass/fail checks in this calculation.

## 7. Applicability

WARNING — record.gamma_G_sup: gamma_G_sup: a tabulated value not taken from a reference record (EN 1990 partial and ψ factors); enter it as one so its source, edition and licence travel with it.
WARNING — record.gamma_G_inf: gamma_G_inf: a tabulated value not taken from a reference record (EN 1990 partial and ψ factors); enter it as one so its source, edition and licence travel with it.
WARNING — record.gamma_Q: gamma_Q: a tabulated value not taken from a reference record (EN 1990 partial and ψ factors); enter it as one so its source, edition and licence travel with it.
WARNING — record.xi: xi: a tabulated value not taken from a reference record (EN 1990 partial and ψ factors); enter it as one so its source, edition and licence travel with it.

## 8. Verification

Template struct.load-combinations@1.0.0 Content hash 5f5b64f9150967c5 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0